

A Droop Quota DAE Model for *Mycobacterium smegmatis* Batch Cultivation Under Nutrient-Replete and Nitrogen-Limited Conditions

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Abstract—Tuberculosis remains a major global health threat, driven in part by the ability of *Mycobacterium tuberculosis* to persist within hostile, nutrient-depleted environments such as granulomas. The non-pathogenic surrogate *Mycobacterium smegmatis* is widely used to study mycobacterial metabolism, but experimentally probing stress conditions is resource-intensive, motivating mathematical modeling as a complementary tool. We evaluate and extend a differential-algebraic equation (DAE) framework for *M. smegmatis* batch cultivation, calibrated through a sensitivity-guided, CasADi/Particle Swarm Optimization pipeline, under a stress condition not previously modeled: severe ammonium deficiency. The Monod-based Díaz model fit the nutrient-replete data but could not accurately simulate nitrogen-starvation dynamics using a single parameter set. reformulating the nitrogen balance around a Droop-type internal quota, which decouples external uptake from instantaneous growth, resolved this limitation: the reformulated model reconciled both regimes using a single parameter set, reducing the global NRMSE to below 0.02 under both conditions: 0.0181 under nutrient-replete conditions and 0.0187 under ammonium-deficient conditions. Correlation and information-criterion analyses further show that parameter identifiability is regime-dependent rather than fixed, with AIC and BIC disagreeing on how many parameters the data support. These results indicate that a physiologically motivated internal-quota extension, rather than further Monod-kinetics recalibration, is sufficient to build a reliable *in silico* platform for mycobacterial nutrient stress, with future work extending the approach to glycerol deficiency and combined carbon–nitrogen limitation.

Index Terms—batch cultivation, differential-algebraic equations, Droop kinetics, *Mycobacterium smegmatis*, nitrogen limitation, parameter calibration, parameter identifiability, Particle Swarm Optimization, sensitivity analysis, tuberculosis.

I. INTRODUCTION

Tuberculosis (TB), despite being controllable and preventable, remains among the top 10 causes of death worldwide [1]. It can occur at any age and is primarily transmitted through the air. According to the 2023 WHO report, approximately 10.7 million people fell ill with tuberculosis. Treatment remains challenging because the organism can enter a metabolically dormant state and withstand prolonged antibiotic exposure [1].

The pathogen responsible for this infection is *Mycobacterium tuberculosis*. This is a strictly aerobic rod-shaped bacterium that grows slowly. *M. tuberculosis* possesses a lipid-rich cell envelope that contributes to its intrinsic resistance to several antimicrobial agents. Errors or deviations in prolonged treatment regimens increase the risk of bacterial persistence and relapse [2].

Macrophages constitute an early line of defense by phagocytosing the bacilli. However, *M. tuberculosis* can survive within macrophages and access intracellular nutrients [3]. Granulomas subsequently form around infected cells, creating a microenvironment characterized by hypoxia, nutrient limitation, and low pH. These conditions can promote bacterial dormancy and persistence within the host [4].

Direct experimental work with *M. tuberculosis* is time-consuming and requires specialized high-containment facilities because of its slow growth and pathogenicity [5]. Instead, the non-pathogenic, fast-growing *Mycobacterium smegmatis*

is used as a substitute [6].

Mathematical modeling of *M. smegmatis* under stress has matured to a point where it can serve as a credible *in silico* platform for tuberculosis research, reducing the need for hazardous and time-consuming experiments with *M. tuberculosis*. Computational models allow systematic exploration of parameter spaces and stress conditions that would be experimentally prohibitive with *M. tuberculosis* [7].

Building on this, several studies have examined mycobacterial behavior under different stress conditions. [2] modeled energy metabolism under carbon-limited, hypoxic conditions, proposing that discrete metabolic rerouting allows redox balance to be maintained under severe stress. [6] modeled carbon-nitrogen-dependent aggregation in *M. smegmatis*, suggesting the well-mixed, homogeneous-batch assumption common to *in silico* simulations can break down under certain C/N imbalances. [8] modeled stress survival using genetic mutants, raising the possibility that viability loss under extreme conditions reflects cellular repair processes absent from standard model structures.

This work builds on the DAE framework developed by [5] and extended by [9] and [10], all of which couple the cultivation environment of *M. smegmatis* with its population and metabolism through a system of differential and algebraic equations. Starting from the mathematical structure of [9], we evaluate its predictive capacity under a stress condition not previously modeled, severe ammonium deficiency, and reformulate the nitrogen balance around a Droop-type internal quota that decouples uptake from instantaneous growth, so that a single, static parameter set describes both nutrient-replete and nitrogen-limited regimes. The performance of this reformulated model, together with the calibration strategy behind it, is analyzed in detail to identify remaining weaknesses and areas for improvement.

II. METHODOLOGY

A. Model Evolution

[5] developed a Monod-based model for *M. smegmatis* covering biomass, glycerol, and carbon dioxide (Apiyo model). [10] revised its charge-balance, self-inhibition, and CO₂ dynamics (De Witt model), and [9] added oxygen dynamics and calibrated the model through sensitivity-driven analysis (Díaz model). The model proposed here continues from the Díaz model.

B. Reformulated Mathematical Model

Dynamic expressions are formulated for the following variables: biomass concentration X (g/L), glycerol C (g/L), external ammonium nitrogen N (g N/L), dissolved oxygen O_2 (g/L), carbon dioxide CO_2 (g/L), and nitrogen quota q (g N/g X).

The model describes the growth and decay of X (g/L), which is governed by the specific growth rate, μ (for parameter units refer to Table I), resulting in the same governing equation as that used in all previous models [9]:

$$\frac{dX}{dt} = (\mu - k_d) \cdot X \quad (1)$$

The dynamic carbon, carbon-dioxide, and dissolved-oxygen balances assume that the corresponding consumption or production rates depend on the specific growth rate and biomass concentration. Unlike the Díaz model [9], the reformulated model includes a maintenance term, m_C , accounting for carbon consumption associated with cellular maintenance rather than biomass formation.

Carbon consumption is modeled as:

$$\frac{dC}{dt} = -\left(\frac{\mu}{Y_{X/C}} + m_C \frac{C}{C + k_C}\right)X \quad (2)$$

with $Y_{X/C}$ being the growth yield coefficient and m_C the specific maintenance coefficient.

CO₂ (g/L) dynamics are modeled considering both production through respiration and consumption through bicarbonate formation:

$$\frac{dCO_2}{dt} = \left(\frac{\mu}{Y_{X/CO_2}} + m_C \frac{C}{C + k_C} Y_{CO_2}^m\right)X - k_{a7} \frac{CO_2}{\frac{10^{-pH}}{k_{a7}} + 1} \quad (3)$$

where Y_{X/CO_2} is the yield coefficient, k_{a7} is the equilibrium constant for the CO₂/bicarbonate system, and $Y_{CO_2}^m$ is a maintenance-related CO₂ production coefficient.

O₂ dynamics depend on gas-liquid transfer and microbial consumption:

$$\frac{dO_2}{dt} = k_L a (O_{sat} - O_2) - \left(\frac{\mu}{Y_{X/O_2}} + m_C \frac{C}{C + k_C} Y_{O_2}^m\right)X \quad (4)$$

where $k_L a$ is the volumetric oxygen transfer coefficient, O_{sat} is the oxygen saturation concentration, Y_{X/O_2} is the yield coefficient, and $Y_{O_2}^m$ is a maintenance-related O₂ consumption coefficient.

The most significant contribution to the model lies in the newly incorporated state variable q , representing the internal nitrogen quota or intracellular nitrogen reserves, based on Droop kinetics [11]. The external ammonium balance is given by:

$$\frac{dN}{dt} = -\rho X + q f_{Nrec} k_d X \quad (5)$$

where N is the external ammonium concentration, ρ is the specific nitrogen uptake rate, k_d is the death rate constant, and f_{Nrec} is an N-recycling factor.

The nitrogen uptake rate ρ is defined as:

$$\rho = \rho_{N,max} \frac{N}{N + k_{Nup}} \frac{q_{max} - q}{q_{max} - q_{min}} \quad (6)$$

where $\rho_{N,max}$ (g N/g X/h) is the maximum uptake rate, k_{Nup} (g N/L) is the half-saturation constant for external nitrogen, q_{min} (g N/g X) is the minimum internal quota, and q_{max} (g N/g X) is the maximum internal quota. This formulation includes Monod saturation on external nitrogen and a limiting term that deactivates transporters when the internal quota approaches its maximum [12], [13].

The Droop growth-limitation factor f_{Nq} [11] is defined as:

$$f_{Nq} = \max \left(1 - \frac{q_{\min}}{q}, 0 \right) \quad (7)$$

The dynamics of the newly incorporated variable q are given by:

$$\frac{dq}{dt} = \rho - \mu q \quad (8)$$

representing the balance between nitrogen uptake and dilution by growth.

The central kinetic expression is the specific growth rate μ :

$$\mu = \mu_{\max} \left(1 - e^{-\frac{t}{t_{\text{lag}}}} \right) \left(\frac{C}{C + k_C} \right) \left(\frac{O_2}{O_2 + k_O} \right) \times \left(1 - \frac{X}{X_{\max}} \right) I_{pH} \cdot f_{Nq} \quad (9)$$

The terms represent: the Monod constants for glycerol k_C and oxygen k_O [12]; biomass self-inhibition against excessive accumulation with $X_{\max}$; a lag phase expression with t_{lag} and $\mu_{\max}$; the Droop-based nitrogen limitation factor f_{Nq} ; and a pH inhibition factor I_{pH} .

The pH inhibition factor is modeled using a two-parameter Gaussian functional form:

$$I_{pH} = \exp \left[- \left(\frac{pH - pH_{\text{opt}}}{pH_w} \right)^2 \right] \quad (10)$$

where pH_{opt} is the optimal growth pH and pH_w is the pH inhibition width. If pH equals pH_{opt} , the factor equals 1 (no inhibition). This simplified formulation provided adequate simulation performance. Although a three-parameter cardinal-pH model is available in the literature, it was not implemented in the present study.

To include the pH of the medium, the charged species in the culture and the metabolic products that dissociate and form hydrogen ions were analyzed. This formulation is based on the pH analysis performed in [10]. Further details are provided in the Supplementary Material. The algebraic charge balance is developed as follows:

$$[H^+] = [OH^-] + [HCO_3^-] + [H_2PO_4^-] + 3[Cit^{3-}] + 2[HCit^{2-}] + [H_2Cit^-] - pH_{alk} - [NH_4^+] \quad (11)$$

The pH is then calculated from $[H^+]$ as follows:

$$pH = -\log_{10}[H^+] \quad (12)$$

Overall, the present reformulated model for *M. smegmatis* corresponds to an index-1 DAE system with 6 differential variables (X, C, N, CO₂, O₂ and q), one algebraic variable (pH) and 19 parameters.

C. Computational Tools and Experimental Data

[9]'s extension of the model also resulted in a CasADi-based automated Python package for DAE systems calibration (DAE-Model-Calibration-and-Analysis) providing a modular and adaptable platform for any DAE system [14]. The present study was performed using this package.

The experimental data used in this study were provided by [5]. Independent calibration and validation datasets were available. Although the models include dynamics for CO₂ and O₂, the lack of experimental data for these variables meant that statistical validation focused on X, C, N, and pH.

D. Parameter Calibration

The model was simulated with defined subsets of parameters selected for calibration within feasible bounds, while the remaining parameters were kept fixed; the package supports running several such parameter-subset combinations, each repeated 50 times to ensure convergence stability.

The optimization objective function aimed to minimize the residual errors (sum of squared errors) between simulated states and experimental data. The algorithm used to minimize the function was Particle Swarm Optimization (PSO) with 50 particles and 200 iterations.

Direct global optimization of the full parameter set with stochastic algorithms such as PSO often leads to identifiability issues, parameter compensation, and convergence to sub-optimal local minima in high-dimensional search spaces. An iterative refinement across several parameter subsets was therefore adopted to systematically isolate and correct structural deviations.

E. Parameter Analysis and Statistical Validation

A study of the local sensitivity of the variables to all model parameters for each PSO parameter solution was then performed as previously done by [9] by perturbing parameters around their nominal values.

Statistical validation was also performed within the package for each solution of calibrated parameters of the PSO optimization.

Metrics were obtained based on a comparison of simulation results and experimental validation data, calculating Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), Normalized Root Mean Squared Error (NRMSE), Akaike Information Criterion (AIC), and Bayesian Information Criterion (BIC), just as in previous cases [9]. A Fisher Information Matrix (FIM) was constructed using the sensitivities of the measured variables and the covariance matrix of the experimental errors. The FIM was also used to calculate parameter *t-values*, with $|t| > 2$ interpreted as evidence of local identifiability.

III. RESULTS

A. Baseline Evaluation of the Modified Framework

As a preliminary diagnostic, the baseline Monod model with the optimized parameter set by [9] was applied to both normal and ammonium starvation validation experimental data. These parameter values are reported in Table I. NRMSE values for each variable with experimental data available and for the global model are reported in Table II. Although the global NRMSE under nutrient-replete conditions was below 0.02, it increased approximately 3.5-fold under ammonium-deficient conditions. This result, together with visual inspection of

the simulated trajectories and experimental data provided in the Supplementary Material, indicated that the model was structurally inadequate under nitrogen-limited conditions.

To address the structural limitations observed in the standard Monod formulation and develop a model capable of simulating both nutrient-replete and ammonium-deficient conditions using the same parameter set, a Droop-type internal nitrogen quota [11] and biomass-maintenance terms were therefore incorporated into the model.

TABLE I
OPTIMIZED PARAMETERS FOR DIFFERENT *M. smegmatis* MODELS

Parameter	Units	Diaz model [9]	Reformulated model
μ_{max}	1/h	0.1335	0.2600 ^a
k_O	g/L	0.0010	0.0001 ^b
k_C	g/L	0.0800	0.0500 ^b
k_N	g/L	0.0100	N/A
k_d	1/h	0.0010	0.0010
t_{lag}	h	7.0000	20.0000 ^b
X_{max}	g/L	1.1537	1.1000 ^b
pH_{LL}	-	4.6000	N/A
pH_{UL}	-	7.4000	N/A
pH_{opt}	-	N/A	6.8000 ^a
pH_w	-	N/A	5.5000
I_{val}	-	3.0000	N/A
$Y_{X/C}$	g/g	0.4840	0.6000 ^a
$Y_{X/N}$	g/g	10.4493	N/A
Y_{X/O_2}	g/g	1.3520	1.3520
Y_{X/CO_2}	g/g	0.3500	0.4000 ^a
m_C	g glyc/(g X · h)	N/A	0.0080
$Y_{CO_2}^m$	g CO_2 /g glyc	N/A	0.0500
$Y_{O_2}^m$	g O_2 /g glyc	N/A	1.0000
$\rho_{N,max}$	g N/(g X · h)	N/A	0.0800
k_{Nup}	g N/L	N/A	0.02
q_{min}	g N/g X	N/A	0.0001
q_{max}	g N/g X	N/A	0.1100
f_{Nrec}	-	N/A	0.5000

^a Re-estimated using PSO.

^b Manually adjusted.

TABLE II
ERROR AND INFORMATION-CRITERION METRICS ACROSS MODELS AND CONDITIONS.

Variable	Diaz model		Reformulated model	
	Normal	Ammonium deficiency	Normal	Ammonium deficiency
X NRMSE	0.0952	0.4251	0.0845	0.1498
C NRMSE	0.0951	0.3805	0.0824	0.0916
N NRMSE	0.1960	0.0677	0.2036	0.0418
pH NRMSE	0.0473	0.2662	0.1069	0.1358
Global NRMSE	0.0197	0.0694	0.0181	0.0187
AIC	—	—	-148.85	-144.60
BIC	—	—	29.53	29.53

B. Sensitivity-Guided Preliminary Parameter Optimization

Initially, a broad screening mapped the parametric space by manually testing parameter values against both normal and ammonium-deficiency trajectories, without active optimization. Sensitivity results identified X_{max} , μ_{max} , t_{lag} , pH_{opt} , $Y_{X/C}$, Y_{X/CO_2} , and q_{max} as the most influential parameters

(detailed results in Supplementary Material). Although k_O and k_C ranked lower in the formal sensitivity screen, both proved empirically influential when varied within realistic bounds; together with t_{lag} and X_{max} , they were manually adjusted to the values in Table I, with several remaining parameters kept at their originally defined values.

Manual iteration then narrowed the calibration set to five critical parameters, μ_{max} , $Y_{X/C}$, Y_{X/CO_2} , pH_{opt} and q_{max} , identifying lower and upper bounds where the fit improved in both conditions (Supplementary Material). Although q_{max} showed sensitivity in the sensitivity graphs, varying it within physiological bounds had minimal impact on the macroscopic dynamics, so optimization efforts were focused on the remaining four parameters.

C. Parameter Calibration and Validation

Parameter calibration was performed for the new reformulated model with 15 parameters fixed at the best values identified through manual screening, while μ_{max} , $Y_{X/C}$, Y_{X/CO_2} , and pH_{opt} were re-estimated. A comprehensive breakdown of the acceptable parameter ranges capable of minimizing the NRMSE for each individual state variable (X , C , N , pH), as well as their localized intersections for both the normal and ammonium-deficient regimes, is provided in Supplementary Material.

By evaluating these intersecting ranges, a highly constrained, "global" boundary set bridging both operational conditions was established. The ideal global ranges were identified as follows: $Y_{X/CO_2} \in [0.310, 0.400]$, $Y_{X/C} \in [0.542, 0.650]$, and $pH_{opt} \in [6.7, 7.3]$. PSO initially estimated μ_{max} , after which its value was fixed at 0.26 h^{-1} during the final optimization of the remaining parameters. The refined parameter values are presented in Table I.

The FIM analysis yielded t-values greater than 2 for all parameters included in the identifiability assessment, suggesting local identifiability. Regarding parameter correlation (matrices for both culture conditions in Supplementary Material), a growth-pH-oxygen group (k_O , μ_{max} , t_{lag} , Y_{X/O_2} , pH_w , pH_{opt}) and a carbon- CO_2 group (k_C , Y_{X/CO_2} , $Y_{CO_2}^m$, X_{max}) are both present, but their strength and the composition of the nitrogen-uptake parameters ($\rho_{N,max}$, k_{Nup} , q_{min}) are markedly condition-dependent: $\rho_{N,max}$ in particular is essentially uncoupled from the growth-related parameters under normal conditions but becomes tightly entangled with them under ammonium deficiency. The most independent parameter is q_{max} under normal conditions and k_d under ammonium deficiency.

The statistical evaluation of this final adjustment highlights a substantial reduction in the global error metrics (Table II). Pooled residuals are not normally distributed under either condition, showing a sharper central peak and heavier tails than a Gaussian (Supplementary Material). Local sensitivities are, in absolute terms, dominated by the unvalidated CO_2 state, which should be kept in mind when reading the aggregate parameter ranking (Supplementary Material).

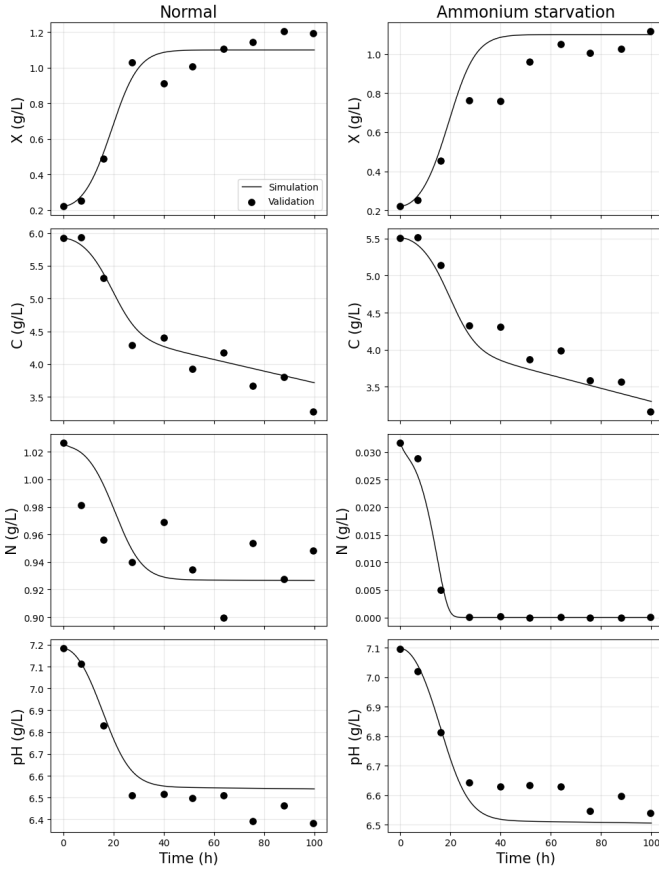


Fig. 1. Simulated trajectories of the reformulated model and experimental data under nutrient-replete conditions (left) and severe ammonium-deficient conditions (right). Dots denote experimental measurements, whereas solid lines denote model simulations.

IV. DISCUSSION

Physiological plausibility of the calibrated parameters. The maintenance coefficient used here (m_C), although not calibrated in this study, is qualitatively consistent with the experimental evidence for carbon expenditure unrelated to growth in stationary-phase *M. smegmatis* reported by [15]. Likewise, the calibrated yield coefficient $Y_{X/C}$ lies above the value used in the normal-condition baseline of [9], consistent with the explicit separation of growth-associated and maintenance-associated carbon consumption in the modified framework: once maintenance is accounted for separately, the fraction of carbon channeled into biomass formation is expected to increase. This separation is also visible in the correlation structure: m_C and $Y_{X/C}$ are tightly linked and largely self-contained under normal conditions, consistent with the two terms being harder to separate when substrate is abundant, but m_C instead tracks the death-rate constant k_d under ammonium deficiency, plausibly reflecting a maintenance demand tied to the stressed, decaying subpopulation once nitrogen becomes scarce.

Comparison with the predecessor oxygen-augmented model. Table II makes the practical payoff of the reformulation

explicit. Under normal conditions the reformulated model is essentially on par with the Díaz model (global NRMSE 0.0181 vs. 0.0197), so the quota extension does not cost accuracy in the regime the predecessor was built for. The gap opens under starvation, where the Díaz model's global NRMSE increased approximately 3.5-fold (0.0197 to 0.0694), driven almost entirely by biomass and glycerol collapsing toward zero as external nitrogen is depleted (X : 0.0952 to 0.4251; C : 0.0951 to 0.3805). The reformulated model instead holds essentially flat across regimes (0.0181 to 0.0187). This near-parity between operating regimes, obtained from a single static parameter set, matters more than the global NRMSE alone: it shows the gain is not achieved by trading starvation accuracy for normal-condition accuracy, but by removing a genuine structural constraint. The trade is not entirely free, as the fits for N and pH worsen slightly (N : 0.1960 to 0.2036; pH : 0.0473 to 0.1069), consistent with a parameter set no longer free to overfit a single regime, which we regard as an acceptable price for cross-regime robustness.

Parameter identifiability and regime-dependence. All parameters in the final set pass the FIM-based t -value threshold ($|t| > 2$), analogous to the identifiability analysis performed by [5]. This is necessary but not sufficient for practical identifiability, however: two parameters can each pass this threshold while being almost perfectly exchangeable with one another. The correlation matrices (Supplementary Material) show this, and, more importantly, show that which parameters are exchangeable is not fixed but regime-dependent: the nitrogen-uptake parameters are essentially uncoupled from the growth- pH -oxygen cluster under normal conditions, but $\rho_{N,max}$ becomes tightly entangled with that cluster under ammonium deficiency, and q_{max} , the most independent parameter under normal conditions, develops new correlations of its own once nitrogen is scarce. Model-selection criteria echo this caution: across the staged calibration reported in the Supplementary Material, BIC favored the simplest four-parameter subset, whereas AIC and NRMSE favored the richer eight-parameter set ultimately adopted. The most defensible reading is kinetic rather than structural: once nitrogen becomes rate-limiting, several specific rates follow the same nitrogen-limited growth trajectory, reducing the independent information available to distinguish growth, uptake, and recycling parameters. Thus, the collinearity that motivated the quota and maintenance terms reappears at the level of the kinetic parameters. This also speaks to the introductory hypotheses of [2], [6] and [8]: reformulating nitrogen kinetics alone, without invoking aggregation, alternative respiratory pathways, or repair-limited viability, was sufficient to reconcile both regimes, suggesting the starvation mismatch was kinetic rather than requiring those additional mechanisms. A profile-likelihood or FIM eigenvalue analysis would be a natural, regime-aware complement to the t -value check in future work [16].

Extrapolation of Droop kinetics to mycobacteria.

Droop kinetics were adopted here as a pragmatic phenomenological approximation for a heterotrophic, non-photosynthetic organism such as *M. smegmatis*, rather than as

a fully mechanistic representation [11]. The strong empirical fit obtained, both in the uncalibrated and calibrated forms, supports the phenomenological validity of this extrapolation for the specific nitrogen-limitation scenario studied, but it does not by itself establish a mechanistic basis for internal nitrogen storage in mycobacteria; this remains an assumption to be tested against direct measurements of intracellular nitrogen pools in future experimental work.

V. CONCLUSIONS

This study extended a DAE model for *Mycobacterium smegmatis* to describe batch growth under both nutrient-replete and severe ammonium-deficient conditions. Evaluation of the standard Monod-based framework showed that it could not reproduce both nutrient-replete and ammonium-deficient dynamics using a single parameter set.

Reformulating the nitrogen balance around an internal quota following Droop kinetics [11] resolved this limitation: the resulting model reconciled both operational regimes with a single, static parameter set, reducing the global NRMSE to below 0.02 (NRMSE of 0.0181 under normal conditions and 0.0187 under ammonium deficiency). This indicates that coupling classical Monod kinetics [12] for carbon and oxygen limitation with an internal nitrogen reserve for nitrogen stress is a promising route toward a unified, reliable *in silico* platform for *M. smegmatis*, without requiring separate parameter sets for each condition. This conclusion should nonetheless be read with the caveat that Droop kinetics were extrapolated here from a phytoplankton context without direct experimental confirmation of an analogous intracellular nitrogen pool in mycobacteria.

Beyond validating this specific extension, the broader outcome of this work is methodological: a sensitivity-driven, sequential calibration strategy can systematically diagnose structural model inadequacies rather than masking them through indiscriminate global optimization. Future work should apply this same strategy to other nutritional stress conditions, in particular severe glycerol deficiency and combined carbon-nitrogen limitation, and extend the framework to mechanistically grounded submodels and multi-regime architectures such as dynamic flux balance analysis, further consolidating this framework as a reliable *in silico* platform for tuberculosis research.

SUPPLEMENTARY MATERIAL

Parameter correlation matrices, residual diagnostics, sensitivity rankings, and the full calibration record are available at the project repository [14].

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